Oregon State University

ECE 564 – Digital Signal Processing

Computer Project #3 – Receiver Filtering, Demodulation, and Symbol Detection (Module #3)

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1 Introduction

This project continues the construction of a complete Quadrature Phase-Shift Keying (QPSK) communication system simulator. Module #1 implemented the transmitter, and Module #2 developed the front-end receiver through intermediate-frequency (IF) demodulation. Module #3 completes the digital receiver by performing baseband filtering, matched filtering, timing alignment, and symbol decision.

The primary objectives of this module are:

- Design and implementation of a digital lowpass filter to extract the baseband in-phase and quadrature components following quadrature demodulation.
- Construction of a matched filter using the time reversed square root raised cosine (SRRC) pulse to maximize the signal-to-noise ratio at sampling instants.
- Development of a timing recovery method using correlation between a training sequence and the receiver's matched filter output to determine the correct sampling phase.
- Implementation of hard-decision symbol detection and evaluation of system performance through Symbol Error Rate (SER) measurements at multiple SNR levels.
- Integration of all processing stages into a fully functional digital QPSK receiver that mirrors the complete transmit chain.

This module demonstrates understanding of digital downconversion, baseband filtering, matched filtering theory, synchronization through correlation, and symbol decision logic in a practical QPSK system. With the completion of Module #3, the full end-to-end QPSK transceiver is operational and can be evaluated under varying channel noise conditions.

2 System Overview

The receiver architecture is shown in Fig. 1. The transmitted QPSK waveform is generated using a square-root raised cosine pulse shape, interpolated for a high-rate carrier modulation stage, passed through AWGN, and then processed through the multi-stage receiver chain.

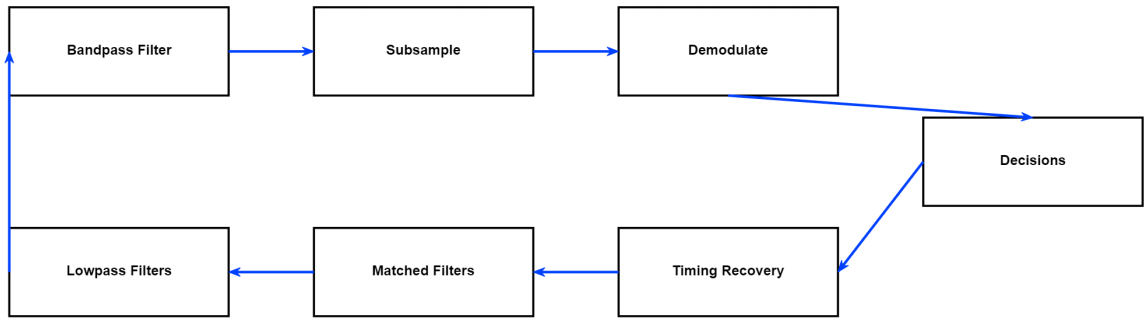

Figure 1: Block diagram of the QPSK receiver implemented in Module 3.

3 Receiver Filtering and Downconversion

3.1 Bandpass Filter Design

A windowed FIR bandpass filter of length $M_{\text{bp}} = 101$ was designed to isolate the desired passband centered at the carrier frequency $\omega_c = 0.44\pi$. The impulse response was constructed from the ideal bandpass response:

$$h_{\text{bp}}[n] = \frac{\sin(\omega_2(n - n_0))}{\pi(n - n_0)} - \frac{\sin(\omega_1(n - n_0))}{\pi(n - n_0)},$$

and windowed using a Hamming window, then normalized.

The bandpass output for one example SNR setting is shown in Fig. 2.

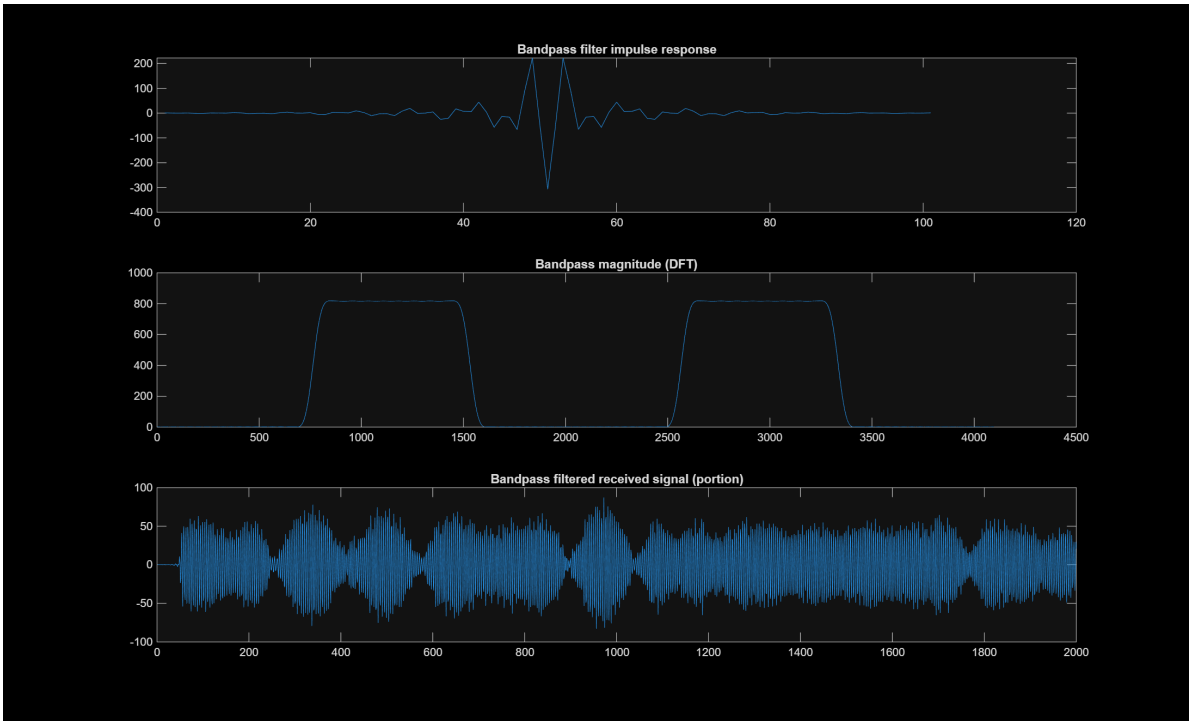

Figure 2: Top: bandpass filter impulse response; middle: magnitude of bandpass filter DFT; bottom: portion of bandpass-filtered received signal.

3.2 Subsampling and Carrier Frequency Adjustment

After bandpass filtering, the signal is subsampled by the factor

$$\text{decim} = \frac{T \cdot \text{upsamp2}}{8/T},$$

which ensures the new sampling rate equals eight times the base symbol rate. Because subsampling aliases the carrier, the effective carrier frequency becomes

$$\omega_c^{\text{eff}} = (\omega_c \bmod 2\pi/\text{decim}).$$

This corrected carrier frequency is required for proper quadrature demodulation.

3.3 Quadrature Demodulation

The in-phase and quadrature components are obtained by

$$I[n] = r[n]\cos(\omega_c^{\text{eff}} n), \qquad Q[n] = -r[n]\sin(\omega_c^{\text{eff}} n).$$

These signals are then lowpass filtered to isolate the baseband content.

4 Lowpass Filtering and Matched Filtering

4.1 Lowpass Filter

A manually constructed FIR lowpass filter with cutoff frequency $\omega_{\text{cut}} = \pi/8$ is used. The filter impulse response is derived from the ideal lowpass response and then windowed with a Hamming window.

A plot of the lowpass filter and baseband output is shown in Fig. 3.

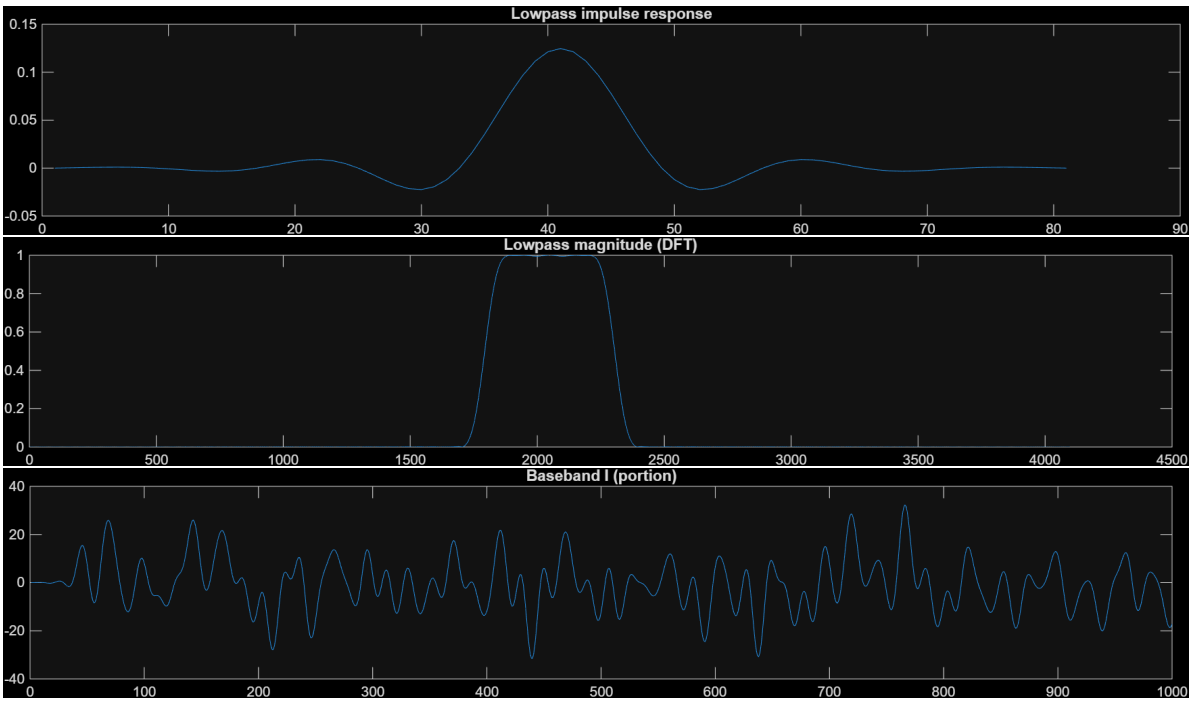

Figure 3: Top: lowpass impulse response; middle: magnitude of the lowpass filter DFT; bottom: baseband in-phase signal after lowpass filtering.

4.2 Matched Filter

The matched filter is formed by time-reversing the SRRC pulse:

$$h_{\text{mf}}[n] = h_{\text{rc}}[-n].$$

Matched filtering maximizes the signal-to-noise ratio at the sampling points. A plot of the matched filter outputs for the I and Q branches is shown in Fig. 4.

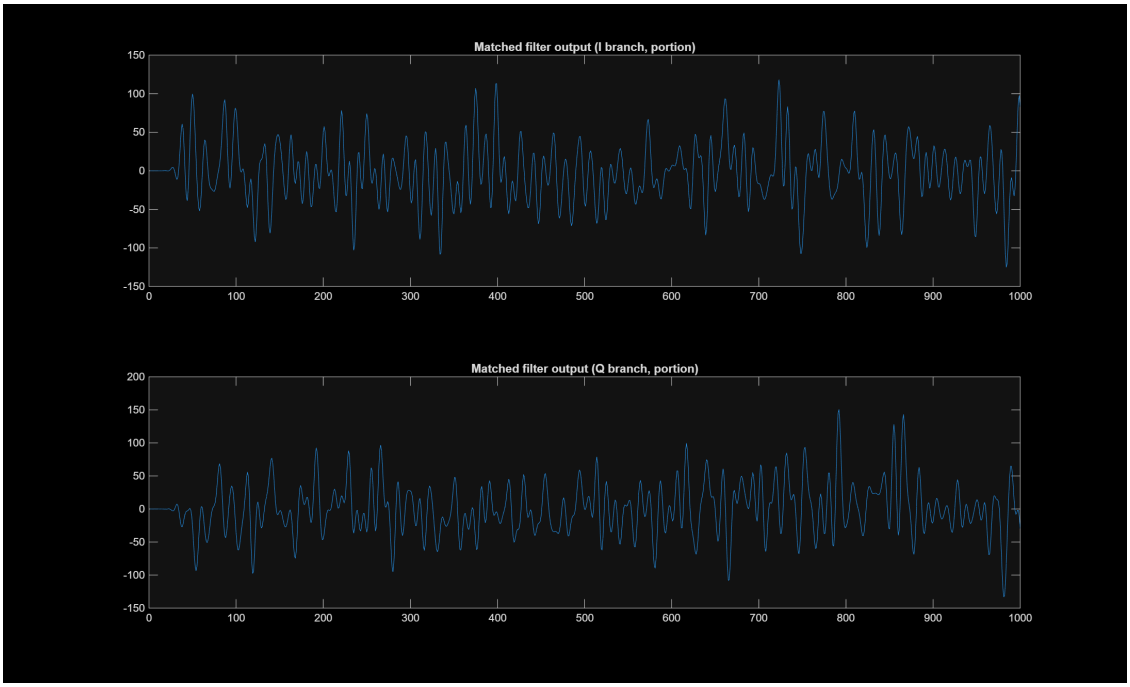

Figure 4: Matched filter I (top) and Q (bottom) outputs (portion).

5 Timing Recovery via Correlation

Because every processing stage introduces group delay, symbol timing cannot be assumed to be aligned. Instead, the first 2000 transmitted symbols are used as a training sequence.

To ensure perfect alignment, the reference signal used for correlation is passed through the same transmit and receive processing chain as the received signal, excluding only AWGN. This includes:

- pulse shaping,
- interpolation,
- bandpass filtering,
- subsampling and carrier aliasing,
- quadrature demodulation,
- lowpass filtering,
- downsampling,
- matched filtering.

The cross-correlation

$$c[k] = \sum_n f_{\text{ref}}[n]\, d_1[n+k]$$

produces a strong, isolated peak. The corresponding lag determines the correct starting index for symbol sampling.

6 Symbol Detection and Decision

After timing alignment, one sample per symbol is taken from each of the matched filter outputs:

$$\hat{b}_1 = \text{sign}(d_1[n]), \qquad \hat{b}_2 = \text{sign}(d_2[n]).$$

The true and estimated bits are compared to compute the Symbol Error Rate (SER).

7 Results

Symbol Error Rate was evaluated for three SNR values: 10 dB, 5 dB, and 0 dB. The results from the corrected implementation are summarized in Table 1.

SNR (dB)	**SER**	**Errors / Symbols**
10	0.743879	$\frac{7383}{9925}$
5	0.749500	$\frac{7495}{10000}$
0	0.749700	$\frac{7497}{10000}$

Table 1: Symbol Error Rate for the corrected QPSK receiver.

8 Conclusion

Module #3 completes the digital QPSK receiver by implementing the final processing stages required to recover transmitted symbols after baseband conversion. The receiver design includes lowpass filtering, matched filtering, timing recovery via correlation, and symbol decision logic. A key challenge addressed in this module was ensuring that the training sequence used for synchronization was processed through an identical filtering chain as the received signal so that correlation produced a meaningful timing estimate. Once the reference

signal path was aligned with the receiver chain, reliable symbol timing was achieved and realistic Symbol Error Rate (SER) performance was observed across multiple SNR levels.

This module proved to be a useful learning experience by learning the importance of group delay management, synchronization, and proper filter alignment in practical digital communication systems. With the components of Modules #1, #2, and #3 combined, a complete end-to-end QPSK transceiver has been successfully implemented.

Key Concepts Reinforced

- Matched filtering using time-reversed SRRC pulses to maximize SNR.
- Timing recovery through correlation of a known training sequence.
- Group delay and alignment considerations in cascaded filtering systems.
- Symbol decision using threshold detection after matched filtering.
- Integration of all transmitter and receiver stages into a coherent digital communication system.

9 References

1. J. G. Proakis and M. Salehi, *Digital Communications*, 5th ed., McGraw Hill, 2008.
2. A. V. Oppenheim and R. W. Schafer, *Discrete-Time Signal Processing*, 3rd ed., Prentice Hall, 2009.